Paleomagnetism of the Plio-Pleistocene Lousetown Formation, Virginia City, Nevada

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The natural remanent magnetization of sixty-one lava flows from the Plio-Pleistocene Lousetown formation was measured. The field area extends from Virginia City, Nevada, to Truckee, California, on the eastern slope of the Sierra Nevada. Geomagnetic polarity epochs and stratigraphic controls are used to determine the general extrusion sequence. The flow-mean directions of magnetization separate into two distinct groups. Thiry-two flows at Lousetown Creek have a $D \simeq 64°$, $I \simeq -67°$, and the remainder of the flows have more usual normal and reverse polarity directions with $D \simeq 20°$, $I \simeq +46°$ (north-seeking directions). Both sets of flows have directions of magnetization that are significantly different from an axial dipole field. It is concluded from the respective directions of magnetization that: (1) the first set of flows represents an intermediate direction of the geomagnetic field during a transition from reverse to normal polarity, and (2) the geomagnetic field deviated significantly from an axial dipole field within the late Pliocene.

INTRODUCTION

The use of paleomagnetic evidence for polar wandering and continental drift hypotheses depends on the assumption that the geomagnetic field has always been a dipole closely aligned with the earth's axis of rotation. Despite numerous applications of paleomagnetic data to these problems, relatively little is known about the variations of the recent paleomagnetic field. Sixty-one flows from the Plio-Pleistocene Lousetown formation were sampled for paleomagnetic analysis. The apparent short time span of extrusion suggests an ideal geologic situation for analysis of short-term variations of the recent paleomagnetic field.

A cursory examination of Figure 7 will reveal considerable variation in the flow-mean directions of magnetization. Since the interpretations of these results depends on the assumption that the paleomagnetic directions represent geomagnetic field directions, the experimental tests of this assumption are described. The interpretations also depend on the age relationships of the various flows sampled; these relationships are considered in the discussion of the paleomagnetic data.

Geologic setting. The Lousetown formation is composed mainly of lava flows but includes a few small intrusive plugs and local fluviatile and lacustrine deposits. The fresh rock of the lava flows is generally a gray basalt or basaltic andesite, although the flows range in composition from olivine latite to basalt [*Birkeland*, 1963].

Following the original description of the type section at Lousetown Creek by *Thayer* [1937], *Thompson* [1956] mapped the flows as a geologic unit in the Virginia City quadrangle, and *Thompson and White* [1964] extended the mapping to the Mt. Rose quadrangle. *Birkeland* [1963] correlated the flows from the Virginia Range across the Carson Range to the eastern slope of the Sierra Nevada (Truckee quadrangle) on the basis of stratigraphic position, topographic projection, and petrologic similarities (Figure 1).

In the Virginia Range the Lousetown flows form two thick sections at Lousetown Creek and Clark Mountain. To the west, in the Carson Range and Truckee basin, the flows cover broad erosional surfaces and generally are single flows with limited areal extent. For the most part the flows are flat-lying or have only a few degrees of dip. Although the major Cenozoic deformation occurred before eruption of the Lousetown flows, the Lousetown formation has several hundred feet of structural relief [*Thompson and White*, 1964].

Several flows in the Truckee basin have been

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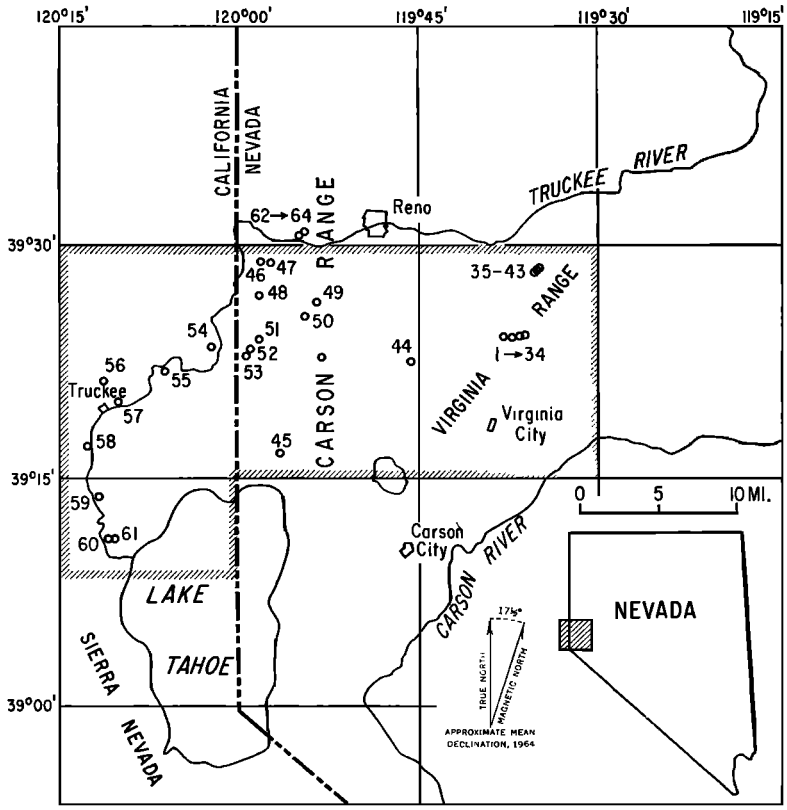


Fig. 1. Index map showing sampling site locations. Flow numbers refer to Table 2.

dated by *Doell et al.* [1966] as part of their study of geomagnetic polarity epochs. The ages of the dated flows range from 1.1 to 1.9 m.y. In the Carson Range the Lousetown flows lie on a post-mature erosion surface cut on the Truckee and pre-Truckee formations, and, as the Truckee formation is Pliocene in age, the Lousetown formation must be Pliocene or younger. A recent K-Ar date on the type section at Lousetown Creek gives an age of 6.8 m.y. (G. B. Dalrymple, personal communication). Thus the Lousetown formation is a Plio-Pleistocene unit.

Field and laboratory methods. A total of 361 cores were drilled from 61 Lousetown flows. Although an attempt was made to sample all the flows in the field area, several small inaccessible outcrops were omitted, and an additional eight flows were deemed unsuitable because of mechanically disturbed outcrops or indications of lightning strikes.

The usual sampling technique consisted of drilling six cores per flow, 2.49 cm in diameter and from 5 to 20 cm in length, with a portable water-cooled diamond coring drill. The samples were oriented by means of a brass mark of known geographic orientation along the length of the core before the sample was broken free from the host rock. In several cases fewer samples were collected, and at four outcrops geographically oriented hand samples were collected and cored later in the laboratory. The majority of the samples from each flow were taken over an approximate distance of 30 to 50 meters with a minimum span of 5 meters and the maximum approximately 1 km. The accuracy of the field orienting method is estimated to be 3°.

For laboratory measurements each core was cut into one or more cylinders 2.30 cm long. Three spinner magnetometers of similar design [*Doell and Cox*, 1965*a*] were used, and the data were reduced graphically and by digital computer. The initial measurements were made at

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the U. S. Geological Survey Rock Magnetics Laboratory, Menlo Park, California, and the later measurements at Stanford University. The direction measurements are reproducible to ±1.1° (s.d.) with an estimated accuracy of ±1.3° (s.d.). Intensity measurements have a precision of ±5% (s.d.) and an estimated accuracy of ±6% (s.d.) [*Heinrichs*, 1966].

The apparatus used for demagnetization experiments is similar to that described by *Doell and Cox* [1963]. It consists essentially of a three-axis tumbler with the current in the coils controlled by a variable transformer. Two specimens per flow were subjected to progressive step demagnetization experiments to determine the optimum field to remove secondary components of magnetization. Duplicate demagnetization experiments and measurements were made on about half the specimens to obtain a better measure of the accuracy of results.

Magnetic Measurements

Original magnetization. The directions of natural remanent magnetization (NRM) for many of the flows, in particular the type section at Lousetown Creek, exhibit a large initial scatter. The extreme scatter of the original magnetization shown in Figure 2 is typical of the section. After demagnetization at 300 oe, however, the directions of magnetization converge to a group. Analysis of the direction changes and corresponding intensity changes during demagnetization suggests that a viscous remanent magnetization (VRM) was removed from specimen 2HO44 and isothermal remanent magnetization (IRM) was removed from specimen 2HO47 (Figure 3).

Before any cores were drilled in the field the outcrops were surveyed with a magnetic gradiometer, and, whenever a deviation of 5° or more was observed, cores were not taken. The initial horizontal directions of magnetization and the large linear intensity decrease on demagnetization suggest that the demagnetization process removed IRM due to lightning [*Cox*, 1961].

Thunderstorms are common in the field area, and, since many Lousetown flows cover topographic highs, they would be particularly susceptible to IRM acquired from lightning. Owing to the lack of strong magnetic field gradients at the sample sites, it is thought that the origin of the scatter is an IRM from old lightning that has partly decayed.

Alternating field demagnetization. Two spec-

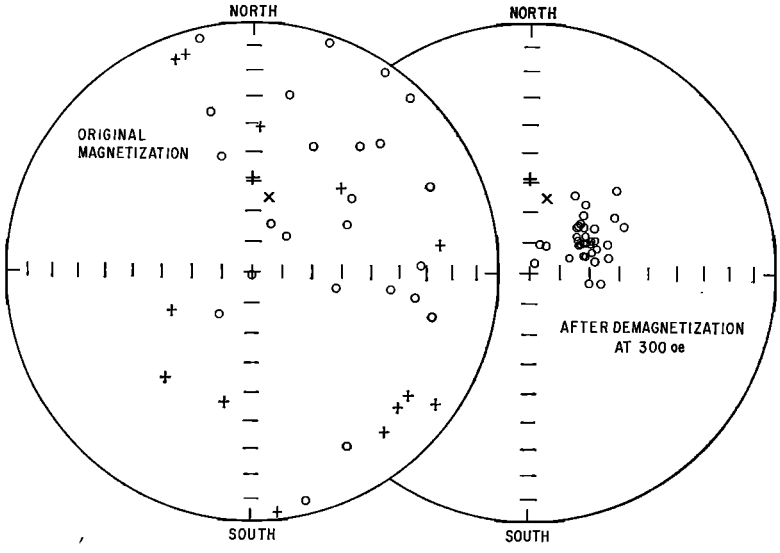


Fig. 2. Directions of remanent magnetization before and after partial demagnetization at 300 oe. Directions represent all specimens from six typical flows of the type Lousetown section. North-seeking direction down, small plus sign; north-seeking direction up, circle; axial dipole field direction, large plus sign; inclined dipole field direction, cross. Lambert equal-area projection.

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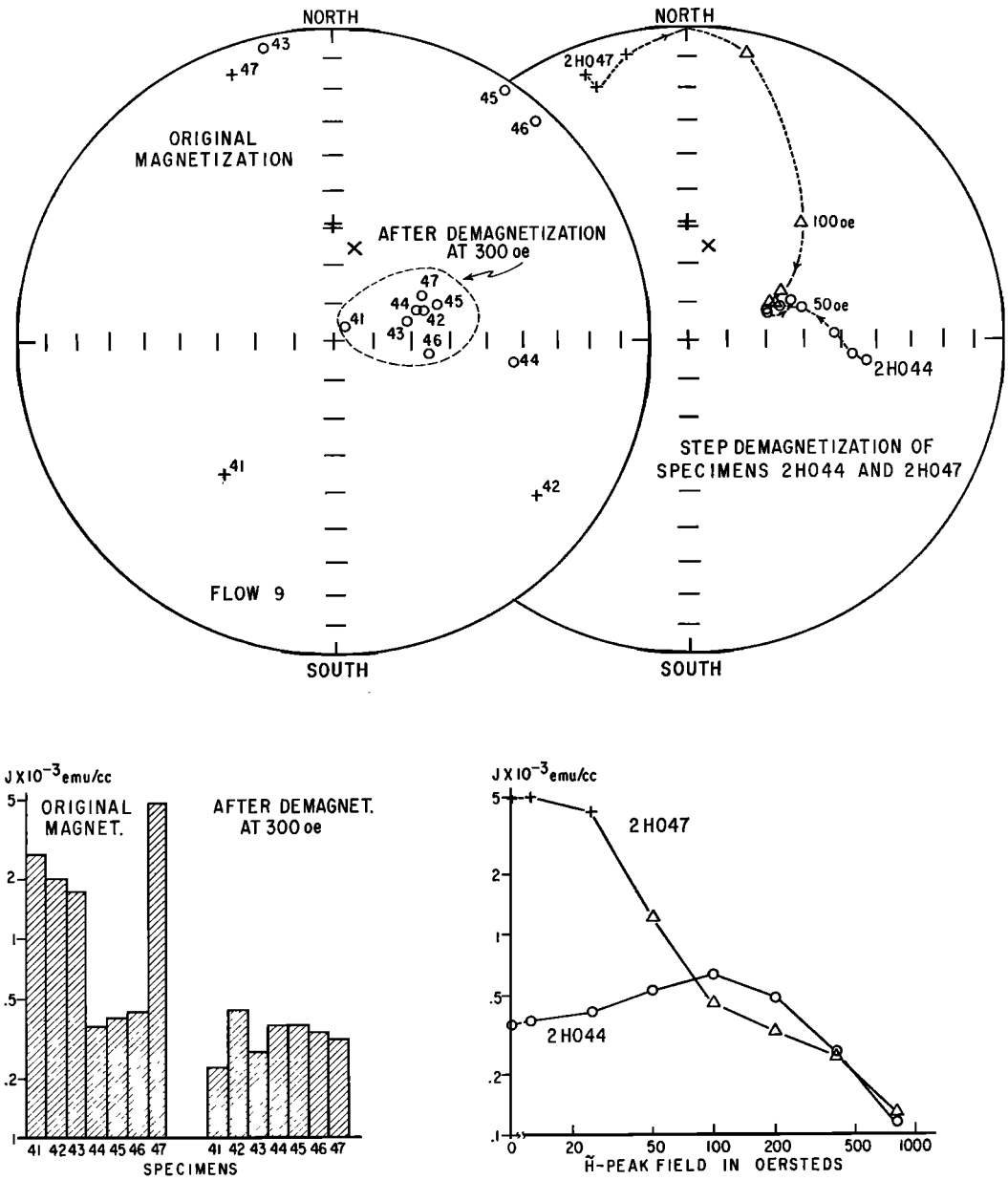


Fig. 3. Summary diagrams for flow 9, showing changes in direction and intensity of remanent magnetization during demagnetization. (See legend for Figure 2.)

imens from each flow were selected for partial demagnetization in alternating magnetic fields to determine the stability of the NRM. To compare apparently stable samples with 'displaced' samples, the first specimen for demagnetization was from within the main group of the original magnetic directions, and the second specimen was from the edge or outside the group. For the Lousetown Creek section with scattered original directions, two widely separated specimens were chosen at random.

Figure 3 shows the two extremes of behavior for the Lousetown Creek section. The plots for specimen 2H047 are typical of most of these

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specimens. Four other representative demagnetization plots are shown in Figure 4. The demagnetization curves for specimens 4HO28 and 4HO31 are quite typical of the flows with moderate initial scatter. The smooth gradual decrease in intensity of magnetization with increasing alternating field intensity together with the systematic change in direction of magnetization of the dispersed samples to a common direction indicates an unstable component of magnetization. This component is selectively destroyed in successively higher fields with the direction of magnetization of the sample moving progressively toward the direction of the more stable TRM component.

For the large majority of the specimens the

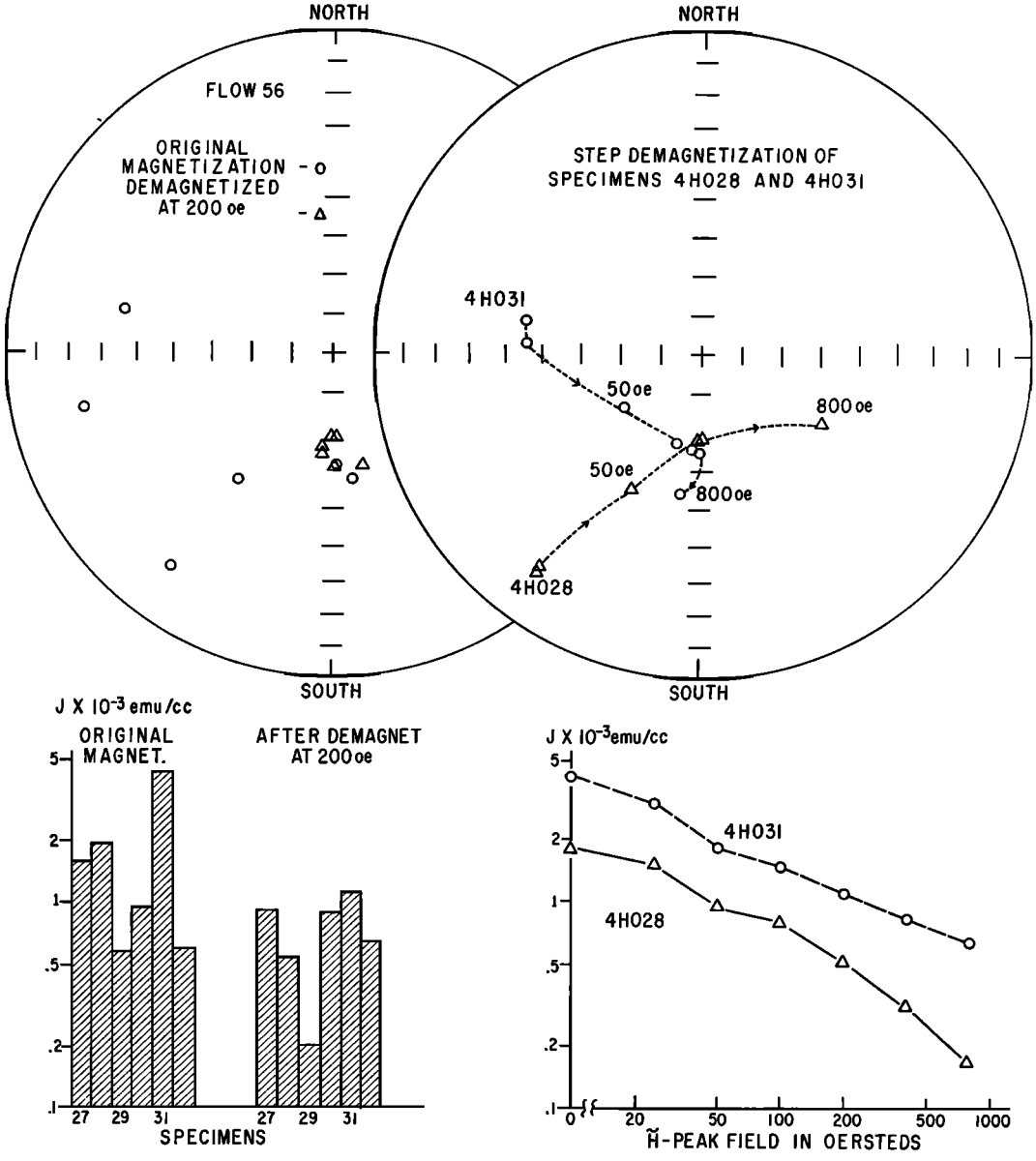


Fig. 4. Summary diagrams for flow 56, showing changes in direction and intensity of remanent magnetization during demagnetization. (See legend for Figure 2.)

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directions converged after demagnetization of 50 to 300 oe and remained grouped in fields up to 600 oe. The remaining specimens from the flow were then demagnetized in the field required to remove the secondary components of magnetization and to reach a stable direction of magnetization. The intensity of the field thus determined varied from 200 to 400 oe (Table 2).

Frequently, large changes in the direction of magnetization occurred after demagnetization in fields of 800 oe (Figure 4). When the experiment is repeated at the same field intensity with the specimen rotated 180° in the demagnetization apparatus, the direction and intensity of magnetization change on successive runs. This indicates that the measured differences are due to an anhysteretic component of magnetization [*Patton and Fitch*, 1962] added to the specimen by the demagnetization apparatus rather than to continued removal of unstable components of magnetization.

Although most of the specimens had consistent within-flow magnetic properties, nine specimens exhibited widely divergent behavior. In eight cases the specimens had displaced directions of magnetization and intensities an order of magnitude larger than the other specimens from the same flow after demagnetization. This suggests that a large IRM component (lightning) was not completely removed by the demagnetization process. The ninth specimen had an unusually small relative intensity of magnetization, and the direction of magnetization continued to change during demagnetization. All nine specimens were excluded from the statistical analyses.

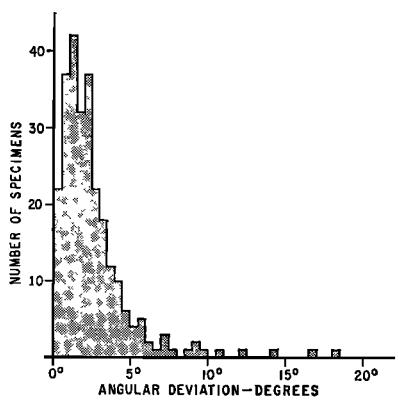


Fig. 5. Histogram of changes in direction of magnetization with repeat measurements.

Storage test. The original magnetization of 263 specimens was measured twice, and ten repeat measurements were made on eleven specimens. Most of the repeat measurements, 155, were made after an approximate time interval of 2 years, and the remainder, 108, were made between 3 days and 6 months apart. The angular differences between repeat measurements are plotted in Figure 5. The angular standard deviation of the entire set is ±2.3°, compared with the precision of the magnetometer itself of ±1.1° (s.d.). This difference is not believed to be significant and is probably due in large part to long-term magnetometer variations. The corresponding intensity changes were minor and less than the precision of the measurements.

As a further check on the time dependence of the measurements the results of the ten repeat measurements were analyzed with respect to the time interval between measurements. Of the eleven specimens, three had large angular differences, 4° to 10°, between successive measurements. The observed changes in direction in 3 days were of the same order of magnitude as the changes in direction observed in 6 months. It appears that these specimens contain a minor magnetic component which is easily realigned by weak magnetic fields. The specimens reach a new equilibrium state very rapidly and do not continue to change. A second observation also supports the conclusion that this variation is not critical. The specimens with large angular differences, 3 days or 2 years, show no tendency to be displaced from the final group when the specimens are demagnetized. Thus on the laboratory time scale no important magnetic instability is present, and observed changes in direction are easily removed by alternating field demagnetization.

Mineralogy. Various investigators [*Sasajima*, 1961; *Kobayashi*, 1959] have shown that chemical changes of the ferromagnetic minerals after a lava flow cools may alter the remanent magnetization. Alternating field demagnetization will not distinguish TRM from a stable chemical remanent magnetization (CRM); both types of magnetization exhibit similar behavior during demagnetization [*Kobayashi*, 1959].

Twenty polished sections were prepared from

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selected Lousetown flows. The particular flows were chosen on the basis of the groupings shown in Figure 9; 10 polished sections from the Lousetown Creek group, 3 from the Clark Mountain group, 3 from the Truckee basin group, and 4 from the Carson Range group. The dominant magnetic mineral in all the flows is a titaniferous magnetite containing exsolved ilmenite. Individual polished sections range from those in which the titanomagnetite has been converted almost completely to hematite and pseudobrookite by high-temperature oxidation to those with no (111) ilmenite lamellas visible except on one or two grains and with maghemite visible on some grains on the margins and along cracks.

Particular emphasis was devoted to the Lousetown Creek section because of the divergent direction of magnetization for the flows (Figure 7). The specimens from the intermediate direction flows include essentially all types of high- and low-temperature oxidation. One might suspect the directions of magnetization of the specimens with only low-temperature oxidation, but the other specimens should give stable directions [*Larson et al.*, 1966]. Since all the directions of magnetization group regardless of the state of oxidation, the intermediate direction probably represents a geomagnetic field direction.

The samples from the other sections exhibit this same range from oxidized to unoxidized states; some of the normally magnetized specimens are oxidized and some are unoxidized, and the same is true for the reversely magnetized specimens. Thus, there does not seem to be any direct correlation between the petrology of the specimens and the magnetic polarity of the flows.

Curie temperatures. Twenty-one Curie temperature measurements were run as an additional check on the magnetic properties of the flows. The apparatus used was a fused quartz torsion balance with the displacement and temperature of the sample displayed on a strip chart recorder [*Clark*, 1965]. Whole rock samples were enclosed in evacuated silica vials and suspended from the torsion beam in a 3500-oe magnetic field. The estimated accuracies for the temperature and saturation magnetization are, respectively, ±10°C and ±8%.

The two major types of heating and cooling curves obtained with the balance are shown in Figure 6. Type 1 is a simple concave curve, indicating a magnetite or titanomagnetite of uniform composition, and type 2 is a more nearly linear curve, without a definite 'break' suggesting titanomagnetites of varying composition [*Akimoto*, 1955; *Doell and Cox*, 1965*b*]. It was not possible to avoid chemical changes completely during heating, but usually these changes were minor, as evidenced by the reproducibility of the heating and cooling curves.

Comparison of the type of Curie temperature curve (Table 1) with the magnetic polarity of the flow finds specimens of both types in all

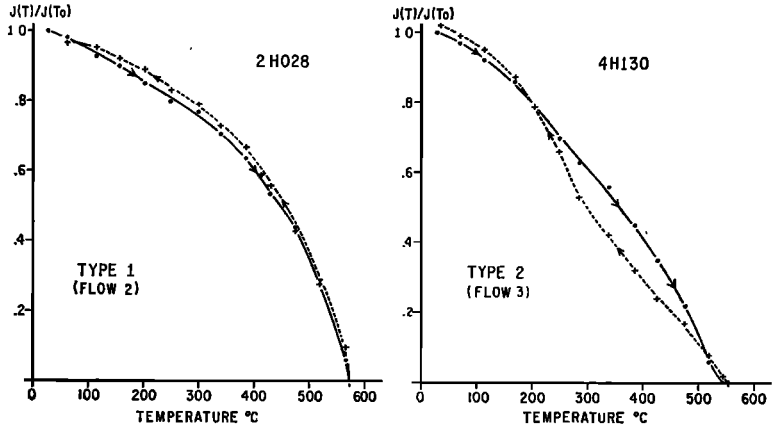


Fig. 6. Examples of saturation magnetization versus temperature curves for the Lousetown lavas. Ordinate is magnetization at temperature T relative to magnetization at room temperature in a magnetic field of 3500 oe. Dots, heating cycle; crosses, cooling cycle.

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TABLE 1. Curie Temperature Measurements

Flow Number	T_c, °C Heat	Cool	Curve*	Polarity†
3	535	555	2	I
6	575	575	1	I
10	530	540	2	I
13	530	550	2	I
16	565	565	1	I
19	575	585	1	I
22	540	555	1	I
25	560	555	1	I
27	505	500	1	I
30	570	570	1	I
34	500	495	2	N
35	380	490	2	R
38	555	545	1	R
41	595	575	1	R
45	575	575	1	R
48	575	575	1	N
50	575	550	1	N
53	575	575 360?	2	R
54	590	590	2	N
57	585	585	1	R
58	560	565 450?	2	?

*Type of thermomagnetic curve as illustrated in Figure 7.

† Magnetic polarity: I, intermediate directions; N, normal polarity; R, reverse polarity.

the magnetic groups. Thus, these data do not indicate any unique mineralogy as a cause either for the intermediate directions of magnetization of the Lousetown Creek flows or for the reversed flows.

The possibility that a flow has been reheated to near its Curie temperature by an overlying flow and then remagnetized on later cooling is not of major concern where the flows were extruded as single flows. The relatively high Curie temperatures, greater than 500°C, indicate that it is also unlikely that the two thick sections, Clark Mountain and Lousetown Creek, have been remagnetized as single units.

Anisotropy. All magnetic minerals have both a shape and a crystallographic anisotropy of susceptibility. In a rock, however, the effects are observed only if either the crystallographic or the shape axes are aligned. A strong anisotropy in magnetic susceptibility tends to rotate the direction of magnetization into the plane of maximum suspectibility. Magnetite has no strong crystallographic anisotropy of susceptibility, and any significant anisotropy from magnetite is due to the deviation in shape of the grains from a sphere [*Uyeda et al.*, 1963]. Since the Curie temperature measurements show that the dominant ferromagnetic mineral is titanomagnetite, the polished sections were examined for preferred orientation of the magnetite grains. The large majority of the magnetite grains are essentially equant and thus do not show any alignment. The elongate grains present reveal no noticeable preferred orientation. Both the equant habit and the lack of consistent orientation of the elongate grains minimize the development of a strong anisotropy of susceptibility with its consequent deviation of the remanent magnetization directions.

Summary. Although the individual experiments described above are not conclusive, all the experiments together suggest:

1. The unstable components of NRM are removed by alternating field demagnetization.
2. The NRM measured after demagnetization is predominantly TRM acquired when the flows cooled.
3. This TRM is essentially parallel to the direction of the geomagnetic field at the time the lavas cooled.

Results

Flow-Average Magnetization

The average direction of NRM for each flow is plotted in Figure 7. Each flow average is based on the measurement of one specimen from each of the cores collected from the flow after partial demagnetization. The mean directions of magnetization and precision parameters were computed using the methods described by *Fisher* [1953] and are given in Table 2. Three flows (42, 47, 52) have a wide scatter of magnetic directions both before and after demagnetization. The radii of the 95% confidence intervals are 80.9°, 69.2°, and 81.5°, respectively. The cause of the large scatter and low precision for these flows is not known. Owing to the low precision of the results, these flows are not included in subsequent calculations.

Flow 58 was sampled at an outcrop separated from the main body of the mapped flow by the Upper Truckee Canyon [*Birkeland*, 1963]. The

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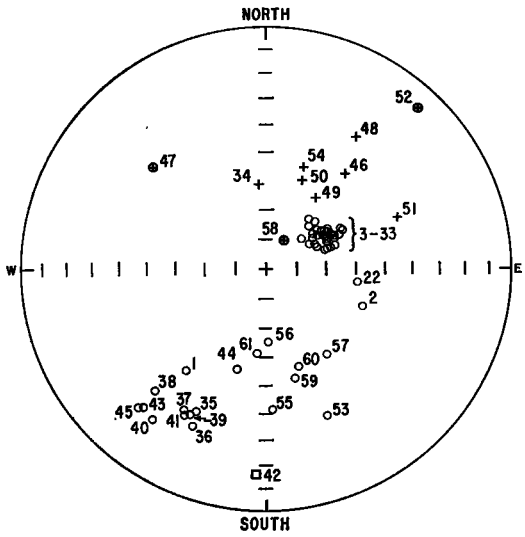


Fig. 7. Average direction of remanent magnetization for each flow after partial demagnetization at 200 or 300 oe. Flow numbers refer to Table 2. North-seeking direction down, plus sign; north-seeking direction up, circles; rejected flows, circles with cross and squares.

intensity of magnetization is anomalously low compared with the other Lousetown flows (Table 2), and the direction of magnetization is inconsistent with results from the same mapped unit by *Doell et al.* [1966]. In contrast to the normal polarity shown here, their results from a site on the main body of the flow indicate a reversed polarity for the flow. It is possible that an underlying pre-Quaternary flow was inadvertently sampled, and flow 58 is not part of the Main Lousetown sequence. Until this uncertainty is resolved by additional sampling, the results from this site will not be included as a reliable Lousetown datum.

The remaining flow-mean directions separate into two distinct groups. Thirty-two of the thirty-four flows from the type section at Lousetown Creek have an unusual intermediate direction ($D \simeq 64°$, $I \simeq -67°$). The other flows, including two at Lousetown Creek, have more usual normal and reverse directions of magnetization ($D \simeq 20°$, $I \simeq +46°$, north-seeking direction). Since any interpretation of

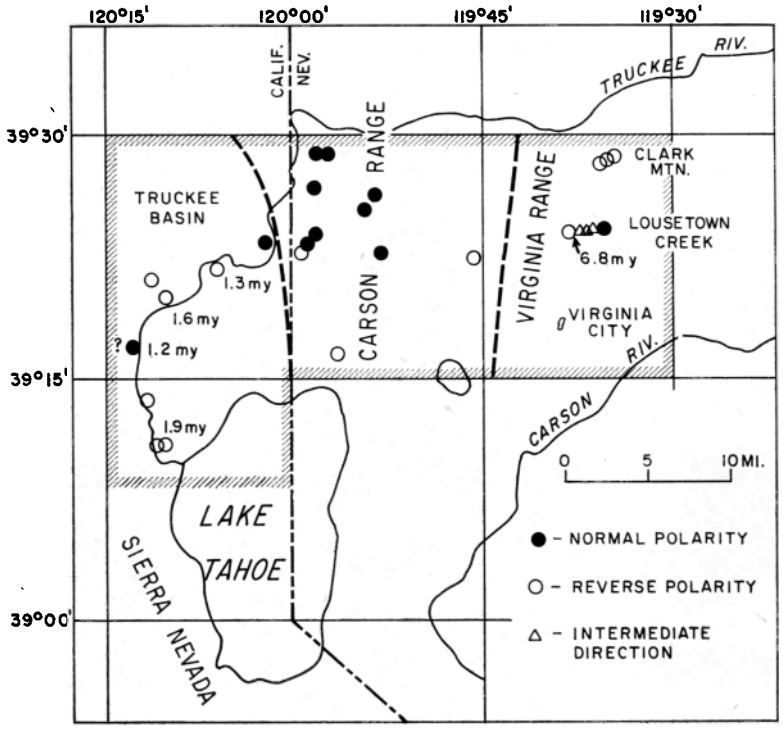


Fig. 8. Index map showing magnetic polarity of individual flows.

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TABLE 2. Flow-Average Directions of Magnetization

Flow	Intensity of Magnetization, oe	Inclination of Flow-Mean Direction	Declination of Flow-Mean Direction	$J_{ave} \pm$ s.d.	Exp.*	Number of Samples	R†	Intraflow Precision Parameter [*Fisher*, 1953]	α_{95}‡	ASD§	Latitude of Virtual Geomagnetic Pole	Longitude of Virtual Geomagnetic Pole
					Virginia Range							
F1	200	−46.4	217.9	4.68 ± 3.25	−4	6	5.765	21.3	14.9	17.6	−56.6	159.2
F2	400	−55.6	111.5	6.65 ± 3.32	−5	6	5.727	18.3	16.1	18.9	−37.1	350.0
F3	300	−67.4	61.2	3.15 ± 1.49	−4	6	5.776	22.4	14.5	17.1	−14.4	25.0
F4	300	−69.6	56.1	3.64 ± 1.03	−4	6	5.983	293.5	3.9	4.7	−14.6	29.6
F5	300	−71.8	62.2	6.31 ± 7.06	−4	6	5.981	260.8	4.2	5.0	−19.4	29.3
F6	300	−66.5	55.7	4.62 ± 0.92	−4	6	5.985	324.3	3.7	4.5	−11.2	26.9
F7	300	−67.1	45.1	2.48 ± 0.56	−4	6	5.876	40.4	10.7	12.7	−7.6	32.9
F8	300	−62.9	64.0	3.48 ± 1.09	−4	7	6.779	27.1	11.8	15.6	−11.6	19.3
F9	300	−69.6	71.7	3.32 ± 0.71	−4	7	6.917	72.7	7.1	9.5	−21.4	22.9
F10	300	−65.0	56.5	2.68 ± 1.46	−4	6	5.840	31.3	12.2	14.5	−9.9	25.1
F11	300	−61.2	60.6	2.70 ± 1.21	−4	6	5.731	18.6	16.0	18.8	−8.4	19.8
F12	300	−73.2	59.8	2.85 ± 0.54	−4	7	6.979	290.4	3.5	4.8	−20.1	32.0
F13	300	−71.3	66.0	3.67 ± 1.58	−4	6	5.981	260.1	4.2	5.0	−20.4	27.2
F14	300	−63.8	64.8	2.45 ± 1.54	−4	6	5.956	112.5	6.3	7.6	−12.8	19.8
F15	300	−67.9	54.5	4.12 ± 2.23	−4	6	5.936	77.7	7.6	9.2	−12.1	28.7
F16	300	−66.3	71.9	5.09 ± 3.35	−4	6	5.936	78.0	7.6	9.2	−18.6	19.0
F17	300	−65.5	57.2	6.42 ± 3.08	−4	5	4.954	86.6	8.3	8.7	−10.8	25.2
F18	300	−67.1	65.0	4.42 ± 1.06	−4	6	5.977	214.3	4.6	5.5	−15.9	23.0
F19	300	−67.6	62.9	4.29 ± 2.13	−4	6	5.991	552.5	2.9	3.4	−15.4	24.5
F20	300	−64.4	64.1	3.06 ± 1.70	−4	6	5.818	27.5	13.0	15.4	−13.0	20.6
F21	300	−68.8	71.3	2.64 ± 0.50	−4	5	4.971	140.2	6.5	6.8	−20.5	22.1
F22	300	−58.5	98.5	1.19 ± 2.49	−3	6	5.753	20.2	15.3	18.0	−29.3	358.8
F23	300	−65.8	71.5	2.50 ± 0.78	−4	7	6.931	87.3	6.5	8.7	−17.9	18.6
F24	300	−71.4	57.8	4.74 ± 2.09	−4	6	5.936	77.9	7.6	9.2	−17.2	30.7
F25	300	−70.2	54.8	3.97 ± 2.13	−4	6	5.913	57.4	8.9	10.7	−14.8	30.8
F26	300	−65.0	62.2	4.93 ± 5.93	−4	6	5.941	84.1	7.3	8.8	−12.6	22.2
F27	300	−64.5	66.8	6.78 ± 1.35	−4	6	5.979	237.3	4.4	5.3	−14.5	19.5
F28	300	−60.8	63.4	8.32 ± 9.59	−4	2	1.992	127.3	22.3	7.2	−9.6	17.9
F29	300	−68.1	52.2	1.89 ± 0.99	−4	6	5.928	69.3	8.1	9.7	−11.4	30.0
F30	300	−69.9	46.1	2.80 ± 1.09	−4	6	5.964	138.9	5.7	6.9	−11.3	34.7
F31	300	−74.7	49.2	2.90 ± 2.21	−4	6	5.969	159.8	5.3	6.4	−18.3	37.9
F32	200	−67.7	41.3	2.17 ± 0.93	−4	5	4.836	24.5	15.8	16.4	−7.1	35.4
F33	200	−67.5	60.4	3.94 ± 1.13	−4	5	4.964	111.4	7.3	7.7	−14.2	25.4

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F34	300	61.1	356.8	1.98 ± 1.72	−3	5	4.977	177.7	5.8	6.1	86.3	199.5
F35	300	−35.3	205.8	2.24 ± 0.87	−4	5	4.811	21.2	17.0	17.6	−60.2	184.9
F36	300	−29.9	205.0	3.89 ± 1.33	−4	6	5.932	73.9	7.8	9.4	−58.0	190.4
F37	300	−33.7	209.8	2.21 ± 2.85	−4	6	5.927	68.3	8.2	9.8	−56.7	181.0
F38	300	−33.1	222.3	5.48 ± 3.08	−4	6	5.888	44.8	10.1	12.1	−47.7	168.4
F39	300	−33.1	207.5	7.54 ± 2.78	−4	6	5.887	44.4	10.2	12.2	−58.0	184.5
F40	300	−24.1	216.8	9.88 ± 4.44	−4	6	5.742	19.4	15.6	18.4	−47.9	179.7
F41	300	−32.1	208.6	2.66 ± 1.21	−4	3	2.994	329.3	6.8	4.5	−56.8	183.9
F42	300	−16.7	182.7	6.20 ± 7.36	−4	6	2.960	1.6	80.9	63.2	...	...
F43	300	−25.3	221.2	6.10 ± 5.73	−4	4	3.683	9.5	31.5	26.4	−45.3	174.6
					Carson Range							
F44	200	−54.2	196.2	7.67 ± 2.36	−4	6	5.859	35.4	11.4	13.6	−70.3	156.3
F45	200	−24.2	222.3	1.52 ± 1.17	−3	6	5.537	10.8	21.4	24.7	−44.2	173.7
F46	200	47.5	40.1	2.98 ± 1.77	−4	5	4.727	14.7	20.7	21.2	55.4	335.8
F47	200	38.2	312.4	5.65 ± 4.98	−4	3	2.530	4.3	69.2	39.3	...	...
F48	200	34.0	34.3	4.53 ± 3.19	−3	6	5.868	38.0	11.0	13.1	53.9	355.2
F49	200	60.1	35.3	5.30 ± 1.69	−3	5	4.980	200.8	5.4	5.7	63.2	315.4
F50	200	57.5	22.7	2.33 ± 0.47	−3	6	5.978	224.1	4.5	5.4	72.3	327.2
F51	200	41.8	68.8	2.90 ± 0.42	−4	5	4.996	1063.8	2.3	2.5	31.0	323.0
F52	200	10.0	43.1	9.18 ± 4.97	−4	4	2.671	2.3	81.5	53.9	...	...
F53	200	−36.1	157.5	3.33 ± 1.36	−3	5	4.764	16.9	19.7	19.7	−62.7	291.5
					Truckee Basin							
F54	200	52.6	20.2	1.59 ± 0.90	−3	6	5.726	18.2	16.1	19.0	72.6	344.6
F55	200	−41.2	177.8	9.04 ± 0.10	−4	5	4.887	35.4	13.0	13.6	−74.2	247.5
F56	200	−66.1	178.5	7.18 ± 3.30	−4	6	5.980	246.5	4.3	5.2	−80.9	53.6
F57	200	−54.7	144.6	8.32 ± 2.96	−4	6	5.858	35.3	11.4	13.6	−61.8	332.8
F58	200	78.2	33.3	5.91 ± 2.63	−5	5	4.892	37.0	12.8	13.3	...	...
F59	200	−51.7	165.7	2.08 ± 2.29	−3	6	5.913	57.2	8.9	10.7	−76.5	303.4
F60	200	−55.2	161.8	1.37 ± 1.14	−3	6	5.950	100.6	6.7	8.1	−75.2	322.0
F61	200	−61.4	185.7	1.40 ± 0.30	−3	2	1.991	112.4	23.8	7.6	−84.5	110.0

* $J_{ave.}$ intensity of magnetization (emu/cc); s.d., standard deviation of intensity; exp., exponent of intensity of magnetization.
† Vector sum of N unit vectors.
‡ Radius of 95% confidence interval about direction of **R**.
§ Angular standard deviation of specimen directions.

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these results depends on the age relationships between the sampled flows, the geological correlations are discussed below.

Geological Correlations

A common problem in attempts to correlate volcanic sequences from one area to another is the lack of adequate evidence useful for stratigraphic correlation. If the flows are 'stacked up,' the relative position of each flow may be determined, but the time interval between flows is still unknown, and exact correlations of different sections are difficult. The occurrence of both normal and reverse polarities of the ancient geomagnetic field is well established. The determination of the chronology of geomagnetic polarity epochs by *Cox et al.* [1964] and *Doell et al.* [1966] for the last four million years enables one to put some restrictions on the age of a young flow depending on its magnetic polarity.

Examination of the flow-mean directions of magnetization reveals several general features (Figures 7 and 8). Reversed magnetic polarities exist both in the Truckee basin and in the Virginia Range, separated geographically by a group of normal polarity flows in the Carson Range. The two groups of reversely magnetized flows have different magnetic declinations and inclinations. The type section at Lousetown Creek has an unusual range of polarity from reversed to intermediate to normal (bottom to top). Each set of flows with distinct directions of magnetization must have been extruded at a different time. The individual sections are discussed below.

Truckee basin. The dated Truckee basin flows (1.1 to 1.9 m.y.) are within the Matuyama reversed polarity epoch (Figure 9). Structural relationships between the flows together with the K-Ar dates indicate that all the reversely magnetized flows were extruded during the Matuyama reversed epoch. The normal polarity flow is probably the oldest Lousetown flow in the Truckee basin and is interpreted as a Gauss normal polarity flow [*Birkeland,* 1963]. It is also possible that this flow was extruded during the Olduvai event within the Matuyama epoch, but lack of specific evidence for such an occurrence indicates that the more general interpretation should be applied.

Carson Range. As evidenced by the degree of erosional dissection, the Carson Range flows are older than late Pleistocene. Thus, it is highly unlikely that the flows with normal polarities are young enough to be in the Brunhes normal epoch (<0.70 m.y.). The erosion surface that most of the flows cover is a late Pliocene or early Pleistocene feature [*Thompson and White,* 1964] and establishes the maximum possible age of the flows. The normal polarity flows were probably extruded during the Gauss normal epoch (2.4 to 3.4 m.y.); the reversely magnetized flows, during the Matuyama epoch. The reversed flow to the south (flow 45) is an exception. This flow is removed from the other flows, and neither the field evidence nor the paleomagnetic data are conclusive in accurately dating it relative to the other flows. On the basis of its magnetic direction alone this flow is tentatively correlated with the Clark Mountain flows (Figure 9).

Clark Mountain. Even though the reversed direction of magnetization for the Clark Mountain flows differ noticeably from the Truckee basin flows, independent geologic evidence is necessary to determine whether one or two polarity epochs are represented by the reversed directions of magnetization. Section-average directions of magnetization with variations of the same order of magnitude may occur within a single epoch [*Doell and Cox,* 1965*b*].

The Clark Mountain flows cannot be definitely assigned to a given polarity epoch on the basis of available geologic controls. The flows are overlain by the Mustang andesite, which is considerably less eroded and deformed than the Lousetown formation [*Thompson and White,* 1964]. Although considerable time appears to have elapsed, the time interval between the formations cannot be determined precisely. The Mustang andesite has a Brunhes normal polarity at Clark Mountain [*Thompson and White,* 1964] and the Clark Mountain Lousetown flows appear to be either early Matuyama (~2.4 m.y.) or late Gilbert (~3.4 m.y.), depending on the magnitude of this time interval. The degree of dissection of the flows is such that they cannot be much older than the Gauss-Gilbert polarity boundary.

Lousetown Creek. In contrast to the relatively young ages of most of the Lousetown formation, flow 2 at Lousetown Creek has been

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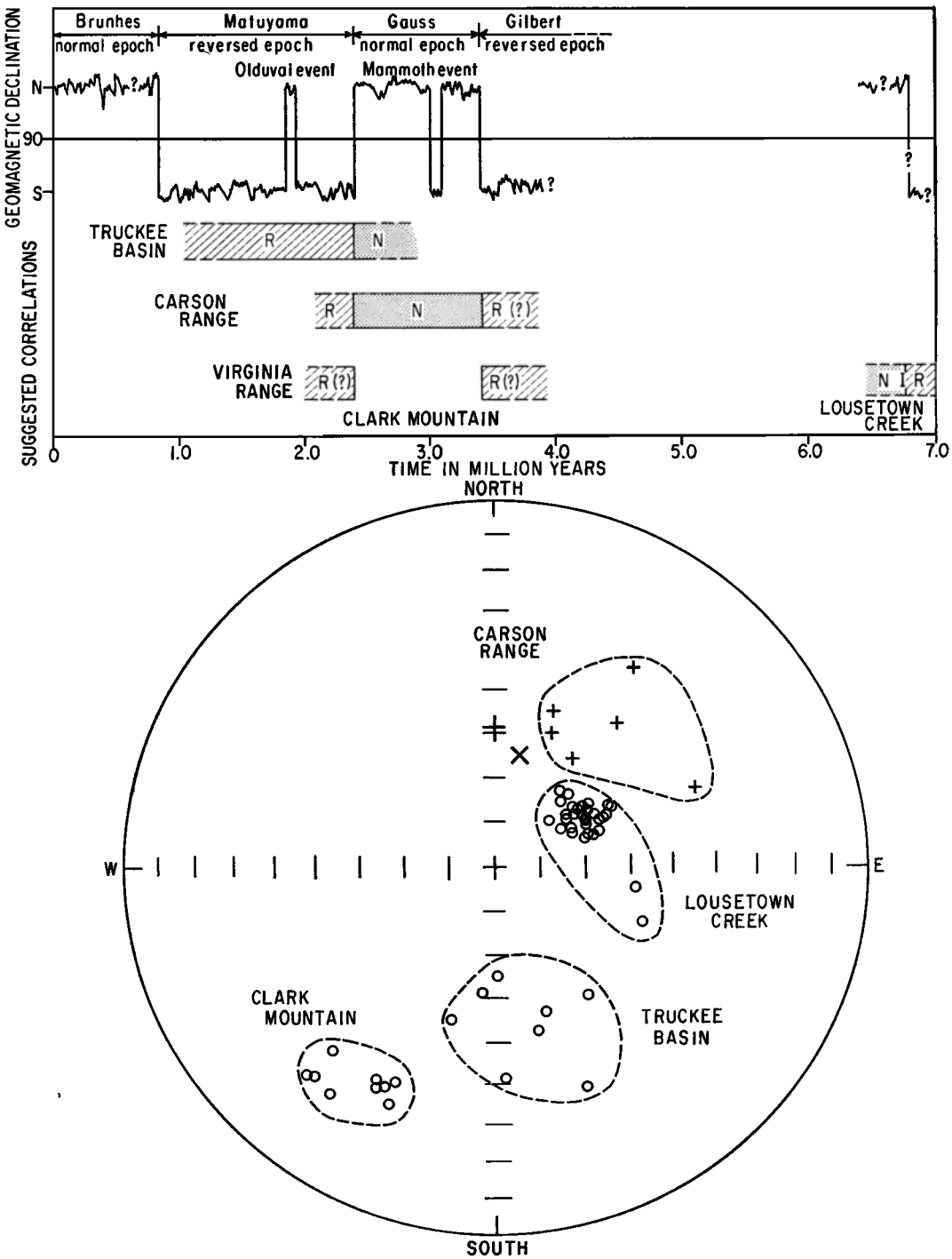


Fig. 9. Suggested correlations of flow-average directions of remanent magnetization with geomagnetic polarity epochs. Polarity epochs after *Doell et al.* [1966]. (See legend for Figure 2.)

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dated at 6.8 m.y. (G. B. Dalrymple, personal communication). Thus, although the Lousetown flows are petrologically similar, the type section of the Lousetown formation is significantly older than the remainder of the Lousetown flows.

Paleomagnetic Analysis

For statistical purposes the flows are treated as four independent populations based on the correlations presented in the preceding section of the paper. All the flows, excluding the Lousetown Creek section, are also treated as a group. The statistical results are summarized in Table 3 and plotted in Figures 10 and 11.

The mean directions of magnetization were computed by means of the methods described by *Fisher* [1953]. The statistical sample must fit three assumptions for the method to be applied rigorously. (1) The data points must be selected at random from a population of unit vectors describing the geomagnetic field. (2) The vectors must be distributed with axial symmetry about their mean direction. (3) The radial density of vectors must decrease with increasing angular distance (ψ) from the mean direction according to the density function

$$P = \frac{k}{4\pi \sinh k} \exp k \cos \psi$$

where k is a precision parameter. The assumptions have not been tested for the Lousetown data since a large number of data points is needed for a definitive statistical test [*Creer et al.*, 1959; *Watson and Irving*, 1957]. Only the section of intermediate directions contains sufficient data for a useful test, and direct observation indicates the first assumption is not satisfied. *Cox* [1964] has shown that, if a large number of random vectors are added to a fixed vector, the resulting distribution closely approximates a Fisherian distribution. If the magnetic variations within each statistical group are random, even though the significance of the

TABLE 3. Statistical Analyses of Flow-Average Directions of Magnetization

DIRECTIONS

Series	Intensity of Demagnetizing Field, oe	Mean Flow Direction: Inclination	Mean Flow Direction: Declination	Number of Samples	R*	κ†	α_{95}‡	ASD§	$J_{ave} \pm$ s.d.¶	Exp.‖
Truckee Basin	200	−53.6	170.0	8	7.794	34.1	9.6	13.9	1.43 ± 0.90	−3
Carson Range	200	50.1	38.0	6	5.825	28.5	12.8	15.2	2.39 ± 2.12	−3
Clark Mountain	300	−30.3	213.4	9	8.927	110 2	4.9	7 7	6.13 ± 4.28	−4
Lousetown Creek	300	−67.4	63.8	32	31.764	131.5	2.2	7.1	4.03 ± 2.13	−4
All flows	···	45.3	22.7	23	21.437	14.1	8 4	21 6	1.36 ± 1.36	−3

VIRTUAL GEOMAGNETIC POLES

Series	Intensity of Demagnetizing Field, oe	Latitude	Longitude	Number of Samples	R*	κ†	α_{95}‡	ASD§	d_m**	d_p**
Truckee Basin	200	80.5	120.2	8	7.673	21.4	12.2	17 5	13.4	9.4
Carson Range	200	58.0	333.7	6	5.775	22.2	14.5	17.2	17 1	11.4
Clark Mountain	300	52.8	359.5	9	8.930	115.1	4.8	7.6	5 5	3 0
Lousetown Creek	300	15.7	203.8	32	31.468	58 2	3.4	10.6	3.7	3.1
All flows	···	67.3	356.8	23	21.044	11.2	9.4	24.2	10.6	6.7

* Vector sum of N unit vectors.
† Between-flow precision parameter estimate [*Fisher*, 1953].
‡ Radius of 95% confidence interval about direction of R.
§ Angular standard deviation of mean flow directions.
¶ Intensity of magnetization (emu/cc) ± standard deviation.
‖ Exponent of intensity of magnetization.
** Semi axes of 95% ovals of confidence.

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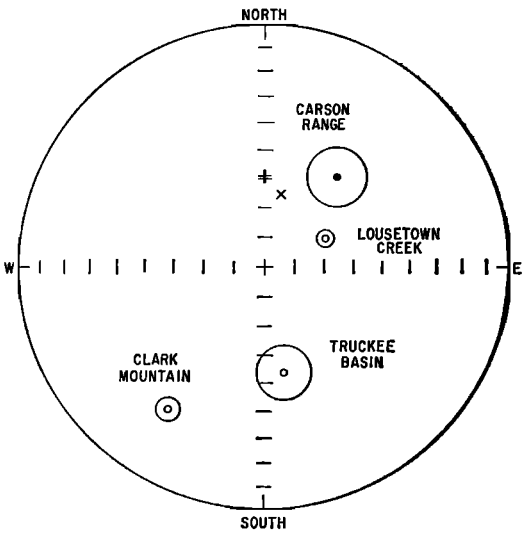


Fig. 10. Section-average directions of magnetization with 95% confidence limits. North-seeking direction down, solid circle; north-seeking direction up, open circle; axial dipole field direction, plus sign; inclined dipole field direction, cross. Lambert equal-area projection.

results cannot be rigorously defined by Fisher's method, Fisher's statistics should yield usable parameters.

Section-average magnetization. Considering only the Truckee basin, Carson Range, and Clark Mountain sections, the directions of magnetization of the Lousetown flows are anomalous in terms of the currently accepted behavior of the geomagnetic field during the Plio-Pleistocene time. To a close approximation the results from other paleomagnetic studies from the late Tertiary and Quaternary show the earth's field to be a geocentric axial dipole [*Irving*, 1964]. The calculations (Table 3) show that the mean virtual geomagnetic pole calculated for the Truckee basin flows is the only pole that is not significantly different from the rotation axis.

The possibility that the section-average directions of magnetization represent non-dipole geomagnetic field directions seems unlikely. The time span of the dated Truckee basin flows is longer than any known periods of non-dipole secular variation. The Carson Range flows resemble the Truckee basin flows both in dispersion of magnetic directions and in general outcrop nature. This strongly suggests that the Carson Range section was extruded over sufficient time to average out the non-dipole secular variation. Since thick sections of lavas can accumulate in a time that is insufficient for non-dipole components to average to an axial dipole, it is possible that the virtual geomagnetic pole for Clark Mountain includes non-dipole components. The good agreement between virtual geomagnetic poles calculated for the normal polarity Carson Range flows and the reversely magnetized Clark Mountain flows suggests, however, that this direction persisted over a reversal of the geomagnetic field and is real. Thus, although the significance of the 95% confidence limits calculated by Fisher's method cannot be rigorously defined, the departure of the paleomagnetic field directions from an axial dipole field appears to be real.

The average displacement of the paleomagnetic field directions from the rotation axis is 22.7°, with individual sections having displacements of 9.5°, 32.0°, and 37.2°. It cannot be determined from the data whether these displacements are due to changes of the rotation axis or to an inclined geomagnetic dipole shifting about a fixed rotation axis.

Twenty-three data points from three (two?) polarity epochs is too small a sample to war-

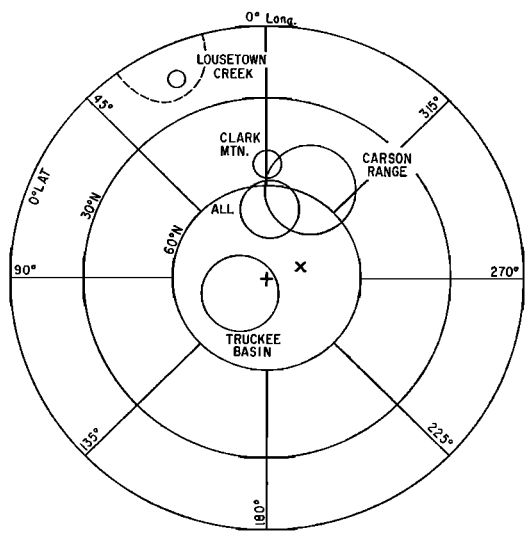


Fig. 11. Section-average virtual geomagnetic poles with 95% confidence limits. All data except Lousetown Creek, ALL; axial dipole, plus sign; inclined dipole, cross. Lambert equal-area projection.

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rant attempts at definitive conclusions about the large scale behavior of the geomagnetic field. The results, however, are consistent with the suggestion that there may be a contribution to the geomagnetic field spectrum with periods of 10^5 to 10^6 years from a random walk of the rotation axis [*Cox and Doell,* 1964]. Before the significance of these directions can be determined, more data are needed from other rocks of the same age.

Intermediate directions of magnetization. The average NRM directions of the Lousetown Creek flows are presented as a function of their stratigraphic position in Figure 12. Three faults in the section are shown by dotted lines. Only in a general sense can the vertical coordinate be treated as a time scale, as neither the total time span of the flows nor the time interval between flows is known. The equal spacing of data points is not intended to imply equal time intervals.

There are no indications of long periods of elapsed time between successive flows. There is only one thin gravel layer in the sequence of thirty-two flows. It is estimated that the flows were extruded over a time interval of from several hundred to possibly several thousand years. Using a conservative estimate of 10 years between each flow, the minimum estimate for the time represented is approximately 300 years.

The most striking feature of the plots is the uniform inclinations and declinations. The total variation in direction is approximately the same as the magnitude of the 95% confidence intervals. The confidence circles for 28 of the 32 flows enclose the mean direction of magnetization of the section. The 4 exceptions are flow 2, the first intermediate flow, flow 12 from the middle of the section, and flows 30 and 31, the last two intermediate flows. The confidence interval of the mean and the confidence intervals of all the flows except flow 2 overlap. Statistically, the flows essentially represent a single direction of the geomagnetic field.

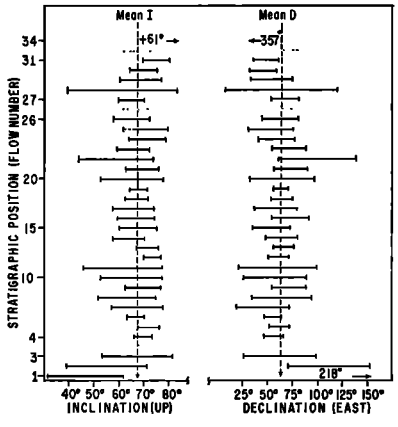


Fig. 12. Declination and inclination versus stratigraphic position for lavas of the Lousetown Creek section. 95% confidence limits.

The intermediate flows crop out in a continuous section with a reversed flow at the base and a normal polarity flow at the top. Thus, it appears that the flows were extruded during a transition of the geomagnetic field from reverse to normal polarity. Although there can be little doubt that the geomagnetic field has reversed polarity many times in the past, relatively little is understood about the behavior of the field during a transition. Two general types of behavior have been postulated. (1) The geomagnetic field retains its predominantly dipolar nature and rotates 180° from one polarity to the other. (2) The main dipole component of the field decreases to zero and then increases again with the opposite polarity. Even though several transition zones have been described [*Sigurgiersson,* 1957; *Byrnjolfsson,* 1957; *Momose,* 1963; *van Zijl et al.,* 1962; *Watkins,* 1965], it is not known which model is correct. The results of this investigation are consistent with either model. The same transition must be found at two widely separated localities before it can be established which general model is correct.

The possibility of an irregular pattern of motion for the geomagnetic field during reversals has been suggested by *van Zijl et al.* [1962] and *Watkins* [1965]. In both cases several consecutive lava flows were found with similar directions of magnetization followed by a large change in direction of magnetization and then another series of closely grouped directions. Two possibilities can explain the results. Either the flows were extruded regularly and the geomagnetic field changed in a series of 'steps,' or the field changed regularly and the flows were extruded in a series of pulses. The large number of Lousetown flows with the same direction of magnetization indicates that the first hypothesis

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is probably true, since the physical number of flows with the same direction of magnetization requires a significant amount of time for eruption.

It has been suggested that the intensity of the geomagnetic field may decrease by a factor of 2 to 5 during a transition [*Momose*, 1963; *van Zijl et al.*, 1962]. The average intensity of magnetization of the intermediate flows after demagnetization at 300 oe is 4.03 ± 2.13 (s.d.) $\times 10^{-4}$ emu/cc in contrast to an average value of 7.91 ± 5.64 (s.d.) $\times 10^{-4}$ emu/cc for 82 specimens from the other flows demagnetized at 300 oe. Both the qualitative nature of this data and the statistical limits indicate the difference shown is not significant.

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